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Design A Chat System

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Figure 1

In this page we first affiliate instant IM software (IMSW) to a standard diagram to indicate the main implementations. For details, we discuss how to design a system that has a scale of one billion users and hundreds of millions of concurrent online users.

Step 1 - Understand the problem and establish design scope

IM is not a single one-on-one chat anymore. It can be a group chat with up to 100 people (Line) or even with up to 500 people (Wechat). Some IMSW (e.g. Slack) even allows third-party integrations. For simplicity, we only consider one-on-one chat and small group chat with no third-party integrations.

Some key features and quantitative requirements are listed below that the interviewer from a hypothetical company asks us to design a system for us. We can add more to the requirements if we think it is necessary during discussions and design and discuss.

Functional requirements:
- IMSW supports 1-on-1 chat and small group chat (up to 100 people).
- IMSW supports online presence indicator.
- IMSW supports multiple devices. The same account can be logged in to multiple accounts at the same time.
- IMSW supports push notification.

Non-functional requirements:
- IMSW is fast. We think of IMSW as real-time messaging.
- IMSW is scalable. We need to handle billions of messages per day.
- IMSW is high availability. The system should not be bogged down by outages.
- IMSW is security. Messages are secure and private.

Scale:
- 1 billion daily active users (DAU)
- 50 billion messages per day
- Assume the average chat length is 100 characters (English)
- Assume the average message size is 1 KB (Text messages only)
- Assume the average message sent per second (QPS) is 100,000
- Assume the peak QPS is five times the average QPS.
- Assume the average memory for caching a user's data is 1 MB.

Storage:
- Assume the average message stored per day is 50 billion
- Assume the average message stored per second (QPS) is 100,000
- Assume the average storage per day is 50 TB
- Assume the average storage per second (QPS) is 50 MB
- Assume the average storage per year is 18 PB

Bandwidth:
- Assume the average bandwidth per second (QPS) is 100 MB

Memory:
- Assume the average memory per second (QPS) is 100 GB
- Assume the average memory per second (QPS) is 1 TB

Hardware:
- Assume the average server has 4 cores CPU, 16GB memory, 512 GB SSD, 10 Gbps network card.
- Assume the average server can handle 10,000 concurrent connections.
- Assume the average server can handle 1,000 QPS.
- Assume the average server can cache 1 million online users' data.
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Figure 2 is a simplified diagram of the system we are designing for. It includes the following components:

- Sender
- Receiver
- Service discovery
- Message service
- Presence service
- Push notification service
- Chat service
- Group service
- Key value store

Figure 2

Where a client sends a chat to another client, the chat comes into one of the stateful servers. For a chat sender, the chat is received by a stateful server. The server then forwards the chat to the receiver.

Figure 3 shows the detailed flow of the messaging system. Service discovery is not shown on the diagram to avoid clutter. We will discuss service discovery in the system design. We will also discuss how to scale the system during the design. Some of the components in the high-level design, for the sender and the receiver, are stateful. The stateful means that the server must be online to receive and process the message. The stateless means that the server does not need to be online to receive and process the message. The stateless means that the server does not need to be online to receive and process the message. The stateless means that the server does not need to